FAHIM DALVI

Software Engineer | Deep Learning Researcher

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EDUCATION

Stanford University

Masters of Science in Computer Science

Artificial Intelligence and Machine Learning

2016 | GPA: 4.0

Carnegie Mellon University

Bachelors of Science in Computer Science

Minor in Mathematics

2014 | GPA: 3.97 (University Honors)

EXPERIENCE

Senior Software Engineer — July 2016 – Ongoing

Qatar Computing Research Institute

- Pre-trained **Fanar** and **Fanar 2.0**, the **best in-class Arabic Large Language Models** on the Open Arabic LLM leaderboard
- Developed award-winning Arabic **Machine Translation** models
- Led the software engineering efforts to develop **toolkits, libraries and scalable APIs** for the translation & interpretation team
- Developed research assets to simplify **interpretation** of deep NLP models
- Co-authored over 40 research papers in areas of machine translation and **explainable AI** in top tier conferences and journals
- Co-taught **Deep Learning** courses to 100+ students

Co-Founder — March 2015 – March 2016

Problemia

Technical lead for designing and developing an **educational platform for teachers**. Managed a team of four and defined the technical direction of the platform

3D Content Creator — Summer 2012

Williams F1

Created **interactive 3D content** for a simulator tuned for Qatar's driving environment

Research Intern — Summer 2012

Robotics Institute, Carnegie Mellon

Designed and implemented a **user interface to analyze results** for a highly distributed data collection system, the AirBoats project

PROJECTS

Fanar

QCRI | 2024 – Present

Pretraining Lead for **Fanar** text models, Qatar's Arabic-centric LLM. Contributed to **modeling, benchmarking** and **deployment** efforts of this National endeavor.

fanar.qcri.org

LLMeBench

QCRI | 2023 – Present

- Created a framework for efficient benchmarking of **Large Language models** across a large suite of tasks
- Cloned over **2000 times** over the course of six months by the community, with over **300k samples evaluated** within QCRI

llmebench.qcri.org

NeuroX: Explainable AI

QCRI | 2018 – Present

- Analyzed and interpreted Machine Translation and **Large Language models** like BERT, RoBERTa, GPT-2 and ALBERT
- Developed a **neural network diagnostic toolkit** for finding and analyzing important neurons for natural language processing models

neurox.qcri.org

Misinformation Detector

QCRI | 2020

Trained and deployed models to detect misinformation in tweets related to the **COVID-19** pandemic

Text-to-Speech

QCRI | 2019 – Present

- Developed a backend and associated REST API to host and run text-to-speech models with **accelerated inference** capabilities
- Created an **ML-Ops pipeline** for seamless deployment of models

tts.qcri.org

Live Speech Translation

QCRI | 2017 - 2020

Designed and developed a live Arabic ↔ English transcription and translation app with a robust backend enabling live broadcastable sessions. Served **100+ hours** of live speech translation sessions

st.qcri.org

Machine Translation API

QCRI | 2016 - Present

Developed a **distributed backend** to manage multiple machine translation engines built at QCRI and a simple to use user-facing REST API, serving over 165 million requests from 40+ countries. Packaged translation engines for **on-premise deployments**.

mt.qcri.org

SKILLS

PyTorch | Keras | Tensorflow | NodeJS | Flask | React | Docker | Python | C++ | Javascript

Deep Learning | Machine Learning | Unix | Web Applications | Backend Development | Scripting | NLP Processing | Containerization | Code & Model Versioning | Testing

TEACHING

Deep Learning for NLP
Carnegie Mellon University
Guest Lecturer | Qatar | October 2022

Deep Learning for NLP
Carnegie Mellon University
Guest Lecturer | Qatar | October 2020

Deep Learning for NLP
University of Duisberg-Essen
Co-Lecturer | Germany | September 2019

Deep Learning for NLP
International Spring School in Advanced Language Engineering (ISSALE)
Co-Lecturer | Sri Lanka | March 2019

Deep Learning for Machine Translation
Computational Linguistics Fall School
Co-Lecturer | Germany | September 2017
Organized by the German Linguistic Society (Deutsche Gesellschaft für Sprachwissenschaft, DGfS).

ACHIEVEMENTS

Best Audience Experience
BBC NewsHack 2017

Best Arabic machine translation system
International Conference on Spoken Language Translation (IWSLT) 2016

Best MYO hack
Hack Overflow, Stanford 2015

Hamad Bin Khalifa University President's Award
Qatar Foundation 2014

Outstanding Academic Achievement
Carnegie Mellon 2014

Senior Student Leadership Award
Carnegie Mellon 2014

1st Place
Oman Collegiate Programming Competition 2012

PATENTS

- "Latent Concept Analysis Method," U.S. Patent No. 12,645,726 B2, issued June 2, 2026. Inventors: Hassan Sajjad, Fahim Dalvi, Firoj Alam, Nadir Durrani, Abdul Rafae Khan, Jia Xu. A method of constructing a dataset for identifying a plurality of latent concepts in a Natural Language Processing model.

PUBLICATIONS

- Mousi, B., Dalvi, F., Chowdhury, S. A., Alam, F., & Durrani, N. (2026a). Said Aloud, Read Different: Cross-Modal Instability in Multimodal Models. *Interspeech 2026*, 3836–3841. https://doi.org/10.21437/Interspeech.2026-1980
- Mousi, B., Dalvi, F., Chowdhury, S. A., Alam, F., & Durrani, N. (2026b). Once correct, still wrong: Counterfactual hallucination in multilingual vision-language models. In M. Liakata, V. P. Moreira, J. Zhang, & D. Jurgens (Eds.), *Findings of the Association for Computational Linguistics: ACL 2026* (pp. 4763–4788). Association for Computational Linguistics. https://doi.org/10.18653/v1/2026.findings-acl.234
- Team, F. (2026). Fanar 2.0: Arabic Generative AI Stack. https://arxiv.org/abs/2603.16397
- Boughorbel, S., Dalvi, F., Durrani, N., & Hawasly, M. (2025). Beyond the leaderboard: Understanding performance disparities in large language models via model diffing. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 31360–31371. https://doi.org/10.18653/v1/2025.emnlp-main.1598
- Durrani, N., Mousi, B., & Dalvi, F. (2025). Editing across languages: A survey of multilingual knowledge editing. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 15906–15918. https://doi.org/10.18653/v1/2025.emnlp-main.803
- Mousi, B., Durrani, N., & Dalvi, F. (2025). An exploration of knowledge editing for Arabic. *Proceedings of The Third Arabic Natural Language Processing Conference*, 417–424. https://doi.org/10.18653/v1/2025.arabicnlp-main.34
- Ersoy, A., Mousi, B. A., Chowdhury, S. A., Alam, F., Dalvi, F. I., & Durrani, N. (2025). From Words to Waves: Analyzing Concept Formation in Speech and Text-Based Foundation Models. *Interspeech 2025*, 241–245. https://doi.org/10.21437/Interspeech.2025-2180
- Team, F. (2025). Fanar: An arabic-centric multimodal generative ai platform. https://arxiv.org/abs/2501.13944
- Mousi, B., Durrani, N., Ahmad, F., Hasan, M. A., Hasanain, M., Kabbani, T., Dalvi, F., Chowdhury, S. A., & Alam, F. (2025). AraDiCE: Benchmarks for dialectal and cultural capabilities in LLMs. In O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, & S. Schockaert (Eds.), *Proceedings of the 31st international conference on computational linguistics*

(pp. 4186–4218). Association for Computational Linguistics. https://aclanthology.org/2025.coling-main.283/

- Yu, X., Dalvi, F., Durrani, N., Nouri, M., & Sajjad, H. (2024). Latent concept-based explanation of NLP models. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 conference on empirical methods in natural language processing* (pp. 12435–12459). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.692
- Mousi, B., Durrani, N., Dalvi, F., Hawasly, M., & Abdelali, A. (2024). Exploring alignment in shared cross-lingual spaces. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 6326–6348). Association for Computational Linguistics. https://aclanthology.org/2024.acl-long.344
- Dalvi, F., Hasanain, M., Boughorbel, S., Mousi, B., Abdaljalil, S., Nazar, N., Abdelali, A., Chowdhury, S. A., Mubarak, H., Ali, A., Hawasly, M., Durrani, N., & Alam, F. (2024). LLMeBench: A flexible framework for accelerating llms benchmarking.
- Hawasly, M., Dalvi, F., & Durrani, N. (2024). Scaled-up discovery of latent concepts in deep NLP models. *18th Conference of the European Chapter of the Association for Computational Linguistics*. https://openreview.net/forum?id=mdMafLCRNg
- Abdelali, A., Mubarak, H., Chowdhury, S. A., Hasanain, M., Mousi, B., Boughorbel, S., Abdaljalil, S., Kheir, Y. E., Izham, D., Dalvi, F., Hawasly, M., Nazar, N., Elshahawy, Y. I., Ali, A., Durrani, N., Milic-Frayling, N., & Alam, F. (2024). LArabench: Benchmarking arabic AI with large language models. *18th Conference of the European Chapter of the Association for Computational Linguistics*. https://openreview.net/forum?id=ssQdvBLc5x
- Fan, Y., Dalvi, F., Durrani, N., & Sajjad, H. (2023). Evaluating neuron interpretation methods of NLP models. *Thirty-seventh Conference on Neural Information Processing Systems*. https://openreview.net/forum?id=YiwMpyMdPX
- Mousi, B., Durrani, N., & Dalvi, F. (2023). Can LLMs facilitate interpretation of pre-trained language models? In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 3248–3268). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.emnlp-main.196
- Dalvi, F., Sajjad, H., & Durrani, N. (2023). NeuroX library for neuron analysis of deep NLP models. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 226–234. https://doi.org/10.18653/v1/2023.acl-demo.21
- Dalvi, F., Durrani, N., Sajjad, H., Jaban, T., Husaini, M., & Abbas, U. (2023). NxPlain: A web-based tool for discovery of latent concepts. *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 75–83. https://doi.org/10.18653/v1/2023.eacl-demo.10
- Alam, F., Dalvi, F., Durrani, N., Sajjad, H., Khan, A. R., & Xu, J. (2023). ConceptX: A framework for latent concept analysis. *Proceedings of the AAAI Conference on Artificial Intelligence*, *37*(13), 16395–16397. https://doi.org/10.1609/aaai.v37i13.27057
- Sajjad, H., Dalvi, F., Durrani, N., & Nakov, P. (2023). On the effect of dropping layers of pre-trained transformer models. *Computer Speech & Language*, *77*, 101429. https://doi.org/https://doi.org/10.1016/j.csl.2022.101429
- Durrani, N., Sajjad, H., Dalvi, F., & Alam, F. (2022). On the transformation of latent space in fine-tuned NLP models. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 1495–1516. https://doi.org/10.18653/v1/2022.emnlp-main.97
- Abdelali, A., Durrani, N., Dalvi, F., & Sajjad, H. (2022). Post-hoc analysis of Arabic transformer models. *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, 91–103. https://doi.org/10.18653/v1/2022.blackboxnlp-1.8
- Abdelali, A., Durrani, N., Demiroglu, C., Dalvi, F., Mubarak, H., & Darwish, K. (2022). NatiQ: An end-to-end text-to-speech system for Arabic. *Proceedings of the The Seventh Arabic Natural Language Processing Workshop (WANLP)*, 394–398. https://doi.org/10.18653/v1/2022.wanlp-1.38
- Sajjad, H., Durrani, N., & Dalvi, F. (2022). Neuron-level interpretation of deep NLP models: A survey. *Transactions of the Association for Computational Linguistics*, *10*, 1285–1303. https://doi.org/10.1162/tacl_a_00519
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- Sajjad, H., Durrani, N., Dalvi, F., Alam, F., Khan, A., & Xu, J. (2022). Analyzing encoded concepts in transformer language models. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 3082–3101. https://doi.org/10.18653/v1/2022.naacl-main.225
- *Dalvi, F., *Khan, A. R., Alam, F., Durrani, N., Xu, J., & Sajjad, H. (2022). Discovering latent concepts learned in BERT. *International Conference on Learning Representations*. https://openreview.net/forum?id=POTMtpYI1xH
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- Alam, F., Shaar, S., Dalvi, F., Sajjad, H., Nikolov, A., Mubarak, H., Da San Martino, G., Abdelali, A., Durrani, N., Darwish, K., Al-Homaid, A., Zaghouani, W., Caselli, T., Danoe, G., Stolk, F., Bruntink, B., & Nakov, P. (2021). Fighting the COVID-19 infodemic: Modeling the perspective of journalists,

fact-checkers, social media platforms, policy makers, and the society. *Findings of the Association for Computational Linguistics: EMNLP 2021*, 611–649. https://aclanthology.org/2021.findings-emnlp.56

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- Sajjad, H., Kokhlikyan, N., Dalvi, F., & Durrani, N. (2021). Fine-grained interpretation and causation analysis in deep NLP models. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Tutorials*, 5–10. https://www.aclweb.org/anthology/2021.naacl-tutorials.2
- Sajjad, H., Abdelali, A., Durrani, N., & Dalvi, F. (2020). AraBench: Benchmarking dialectal Arabic-English machine translation. *Proceedings of the 28th International Conference on Computational Linguistics*, 5094–5107. https://doi.org/10.18653/v1/2020.coling-main.447
- Durrani, N., Sajjad, H., Dalvi, F., & Belinkov, Y. (2020). Analyzing individual neurons in pre-trained language models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 4865–4880. https://doi.org/10.18653/v1/2020.emnlp-main.395
- Dalvi, F., Sajjad, H., Durrani, N., & Belinkov, Y. (2020). Analyzing redundancy in pretrained transformer models. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 4908–4926. https://doi.org/10.18653/v1/2020.emnlp-main.398
- Wu, J., Belinkov, Y., Sajjad, H., Durrani, N., Dalvi, F., & Glass, J. (2020). Similarity analysis of contextual word representation models. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4638–4655. https://doi.org/10.18653/v1/2020.acl-main.422
- *Belinkov, Y., *Durrani, N., Dalvi, F., Sajjad, H., & Glass, J. (2020). On the linguistic representational power of neural machine translation models. *Computational Linguistics*, *46*(1), 1–52. https://doi.org/10.1162/coli_a_00367
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- Durrani, N., Dalvi, F., Sajjad, H., Belinkov, Y., & Nakov, P. (2019). One size does not fit all: Comparing NMT representations of different granularities. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 1504–1516. https://doi.org/10.18653/v1/N19-1154
- Bau, A., Belinkov, Y., Sajjad, H., Durrani, N., Dalvi, F., & Glass, J. (2019). Identifying and controlling important neurons in neural machine translation. *International Conference on Learning Representations*. https://openreview.net/forum?id=H1z-PsR5KX
- *Dalvi, F., *Durrani, N., *Sajjad, H., Belinkov, Y., Bau, A., & Glass, J. (2019). What is one grain of sand in the desert? analyzing individual neurons in deep nlp models. *Proceedings of the AAAI Conference on Artificial Intelligence*, *33*(01), 6309–6317. https://doi.org/10.1609/aaai.v33i01.33016309
- Dalvi, F., Nortonsmith, A., Bau, A., Belinkov, Y., Sajjad, H., Durrani, N., & Glass, J. (2019). NeuroX: A toolkit for analyzing individual neurons in neural networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, *33*(01), 9851–9852. https://doi.org/10.1609/aaai.v33i01.33019851
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- Dalvi, F., *Durrani, N., Sajjad, H., & Vogel, S. (2018). Incremental decoding and training methods for simultaneous translation in neural machine translation. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, 493–499. https://doi.org/10.18653/v1/N18-2079
- Barrón-Cedeño, A., Da San Martino, G., Zhang, Y., Ali, A. M., & Dalvi, F. (2018). Qlusty: Quick and dirty generation of event videos from written media coverage. *NewsIR@ECIR*, *2079*, 27–32.
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- Sajjad, H., Dalvi, F., Durrani, N., Abdelali, A., Belinkov, Y., & Vogel, S. (2017). Challenging language-dependent segmentation for Arabic: An application to machine translation and part-of-speech tagging.

Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 601–607. https://doi.org/10.18653/v1/P17-2095

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